(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

FIFTH SEMESTER (B.TECH) JANUARY-2024

Paper Code: CIC-307 ***Subject: Computer Networks***

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including question no.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt **(Any Five)** **(3x5=15)**

a) What is the purpose of a switch in a network?
b) Explain the difference between half-duplex and full-duplex communication.
c) Explain the purpose of a subnet mask in IP networking.
d) Define the role of a DNS server in network communication.
e) Define the term "fiber optic cable" and explain its advantages in data transmission.
f) How flow and error control works in data link layer?

UNIT-I

Q2 a) What is OSI model. Explain its layers in detail with the help of diagram. Also list the difference between OSI and TCP/IP model? **(10)**

b) An 8 bit byte binary value 10101111 is to be encoded using an even parity Hamming code. What is the binary value after encoding? **(5)**

Q3 a) Explain different types of Guided and Unguided transmission media with the help of diagram. Also list out the advantages and disadvantages of each? **(10)**

b) What is Packet Switching? Explain Virtual circuit and Datagram approach in detail? **(5)**

UNIT-II

Q4 a) What are the various design issues in Data Link Layer? Explain the Stop and Wait ARQ protocol with diagram. **(8)**

b) Compare and contrast HDLC and PPP. Explain the frame format of both with a diagram. **(7)**

Q5 a) Compare and contrast the features of repeaters, hubs, switches, and bridges. Describe the functions of each device and emphasize how each one manages traffic flow and improves network performance. **(8)**

b) Explain channel allocation problem and its solution. Provide an example to illustrate. **(7)**

UNIT-III

Q6 a) What is the difference between classful and classless IP addressing? Explain using an example. **(5)**

b) Explain the Leaky bucket algorithm and illustrate how traffic congestion can be reduced. **(5)**

c) What is the maximum number of subnets in each case? **(5)**

i) Class A; mask 255.255.192.0
ii) Class B; mask 255.255.192.0
iii) Class C; mask 255.255.255.192
iv) Class D; mask 255.255.255.240

P.T.O.

CIC-307

Q7 a) What are distance-vector and link-state routing algorithms. Explain? **(5)**

b) Describe the role of ARP in computer networks. How does ARP help devices communicate within the same network? Provide a basic explanation of the ARP process. **(5)**

c) An organisation is granted the block 211.17.180.0/24. The administrator wants to create 32 subnets. **(5)**

i) Find the subnet mask.
ii) Find the number of addresses in each subnet.
iii) Find the first and last addresses in subnet 1.
iv) Find the first and last addresses in subnet 32.

UNIT-IV

Q8 a) Explain the differences between UDP and TCP in the transport layer. Discuss their main characteristics and provide examples where using UDP might be more advantageous than TCP? **(7.5)**

b) Describe the importance of congestion control in the transport layer. How does congestion control contribute to maintaining network stability and ensuring a reliable data transfer process? **(7.5)**

Q9 a) Explain the client-server model in the context of the application layer. Discuss the roles of clients and servers in this model, highlighting how communication is established between them? **(7.5)**

b) Provide an overview of the Domain Name System (DNS). How does DNS work to translate domain names into IP addresses? Explain the role of DNS servers in facilitating web browsing and internet communication? **(7.5)**

****

CIC-307

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SEVENTH SEMESTER (B.TECH) JANUARY-2024

Paper Code: ETIT-401 ***Subject: Advanced Computer Networks***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions including Q. no.1 which is compulsory. Select one question from each unit.

Q1 Answer the following:

(a) What are then services offered by transport layer? **(3)**

(b) Compare distance vector routing with path vector routing? **(3)**

(c) List down the various characteristics of Big Data? **(3)**

(d) What are the different timers used in Routing Information Protocol (RIP) **(3)**

(e) What do you understand by Next Generation Networks? **(2)**

(f) Define count to infinity problem? **(2)**

(g) Draw the UDP Header format? **(2)**

(h) What the following terms stand for: **(3)**
(i)DVMRP (ii) OSPT (iii) SONET

(i) ARP is used to convent the _____address into_____address. **(2)**

(j) What are the benefits of optical networking? **(2)**

UNIT-I

Q2 Differentiate between IPv4 and IPV6? **(12.5)**

Q3 Explain any one routing protocol in detail? **(12.5)**

UNIT-II

Q4 Draw the header format of TCP and explain each field in details? **(12.5)**

Q5 What is congestion? How TCP deals with the congestion in the network? **(12.5)**

UNIT-III

Q6 Discuss the SONET architecture in details? **(12.5)**

Q7 Discuss the queue management algorithms used by SONET details? **(12.5)**

UNIT-IV

Q8 Define cloud computing? Discuss the various service model supported by the cloud computing? **(12.5)**

Q9 Write short note on the following: **(12.5)**
(a) VolP
(b) SMTP

****

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH.] JULY 2023

Paper Code: ETEC-310 ***Subject: Data communication and Networks***

(IT/ECE/MAE/ITE)

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt any five questions including Q.No1 which is compulsory. Select one question from each unit.

Q1 Attempt **any Five** **[5x5=25]**

(a) Explain the correlation between the OSI model and TCP/IP model with a proper diagrammatical form.

(b) What do you understand by Circuit Switching and Packet Switching? Discuss through the diagram.

(c) Describe the LAN-based FDDI over a basic token ring in detail.

(d) Differentiate SMTP & POP protocol.

(e) Explain the IPv4 protocol with header format. Also, compare with IPv6.

(f) Explain the High-Level Data Link Control (HDLC) protocol used in computer networks. Discuss its key features, frame structure, and modes of operation.

UNIT-I

Q2 a) What is the OSI reference model? Draw a block diagram and explain the functioning of each layer. **[7.5]**

b) i. Define the transmission modes with a diagram. **[2.5]**

ii. Describe the transmission impairment. **[2.5]**

Q3 Explain the following terms with proper examples: **(2.5x5=12.5)**

a) Virtual Circuit with diagram

b) Guided and Unguided Media

c) Wavelength

d) Differentiate Process to Process and Host to Host delivery

e) Baseband transmission and Broadband transmission

UNIT-II

Q4 a) Explain the working of the Go-Back-N ARQ protocol and the Selective Repeat ARQ protocol. Also, compare the Sender and Receiver Window Sizes of both ARQ. **[6.5]**

b) Short note on: **[6]**

i) Leaky bucket algorithm

ii) Channelization in Medium Access Control

Q5 a) Define the meaning of carrier sense in CSMA. Also, Explain the CSMA/CD and CSMA/CA with the diagram. **[6.5]**

b) Explain the following: **[6]**

i. Control Protocols in Medium Access Control

ii. Pure ALOHA and Slotted ALOHA

P.T.O.

(Please write your Exam Roll No.) 1713/23 Exam Roll No. .....

END TERM EXAMINATION

FIRST SEMESTER [B.TECH.] MARCH 2023

Paper Code: ETCS-111 ***Subject: Fundamentals of Computing***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Short answers type questions. **(2.5x10=25)**

a) What do you mean by Protocol?
b) What are external commands? List out any two.
c) What are the responsibilities of Loader, Linker and Assembler in the compiler environment?
d) What is Macro in Libre writer?
e) Differentiate between CUI and GUI.
f) Differentiate between volatile and nonvolatile memory.
g) Explain the MAN and WAN in computer networks.
h) What do you understand by FAT and NTFS?
i) An Education institutional Lab has a fully connected mesh network consisting of 10 devices. Calculate the total number of cable links needed and the number of ports for each device.
j) Draw a flow chart to compute factorial of N?

UNIT-I

Q2 a) Draw a block diagram to illustrate the basic architecture of a computer system and explain the functions of the various units. **(6.5)**

b) Explain in brief dot matrix printer. Differentiate between impact printer and non-impact printer. **(6)**

Q3 a) Draw memory hierarchy diagram and explain all of the levels. **(6.5)**

b) Explain various storages devices of computer in detail. **(6)**

UNIT-II

Q4 a) Write a short note on the following. **(6.5)**
i) Linux architecture
ii) Window System Utilities
iii) Linux File System

b) Explain the following Linux commands in detail. **(6)**
i) PWD ii) RM iii) CHMOD iv) MV iv) CAT vi) VI

Q5 a) Explain Multiprogramming, Multiuser and Multiprocessing with example. **(6.5)**

b) What is DOS? Explain Process management done by operating system. **(6)**

P.T.O.

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SEVENTH SEMESTER [B.TECH] FEBRUARY 2023

Paper Code: ETIT-401 ***Subject: Advanced Computer Networks***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions including Q.No1 which is compulsory. Select one question from each unit.

Q1 Attempt **any Five** of the following:- **[5x5=25]**

a) What are the differences between ARP and RARP?

b) Describe Congestion Control techniques with help of suitable example?

c) What is the difference between IGMP and ICMP, and how they are used in network communication? Describe with the help of suitable diagram.

d) How does the architecture of IPv4 differ from that of IPv6?

e) What is Big Data, and how does it differ from traditional data management approaches?

f) What is error control at the transport layer, and why is it important in data communication?

UNIT-I

Q2 a) Explain the different address spaces in classful addressing? **(6)**

b) For a given IP address 172.16.10.22 and mask 255.255.255.240, find the following: Subnet mask, broadcast address and valid range of IP addresses in this network? **(6.5)**

Q3 Discuss OSPF, RIP, and BGP routing protocols in detail. Explain the functioning of each routing protocol with example. **(12.5)**

UNIT-II

Q4 What is the use of transport layer? Explain the TCP protocols with the help of suitable diagram and example. **(12.5)**

Q5 a) What are the key characteristics of UDP, including its use of data grams, lack of reliability guarantees, and low overhead compared to other protocols? **(6)**

b) Draw the frame format of TCP? Explain how the flow control and error control handled in TCP? **(6.5)**

UNIT-III

Q6 What does SONET stand for? Explain the SONET architecture and its layers in detail. Also, describe the function of the regenerator in detail. **(12.5)**

Q7 a) What is resource management, and why is it important for networks? Explain the different types of resource management in detail. **(6)**

b) Describe QoS for optical networks? Explain Queue Management Algorithms. **(6.5)**

UNIT-IV

Q8 Describe various object oriented testing techniques like class testing, inter class testing and system testing with the help of suitable example. **(12.5)**

Q9 Write down Short note on following (**any two**): **[6.25, 6.25]**

a) SAAS, PAAS and HAAS.

b) Telnet and Finger.

c) Cloud Computing.

****